

LECTURE 12 - THE CHAIN RULE

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ABSTRACT. Last time we introduced partial derivatives, and said that to differentiate a function of multivariables with respect to one variable, we'd treat all of the other variables as constant and use our normal derivative rules. In this lecture we'll explore what happens when we compose functions of multiple variables together, and see what happens when we try to apply the chain rule to differentiate such functions.

1. CHAIN RULE FOR COMPOSITION WITH A SINGLE-VARIABLE FUNCTION

Before we dive into the chain rule for multivariable functions, let's quickly recall the chain rule for functions of a single variable. If $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are two differentiable functions, the chain rule for functions of a single variable tells us

$$\frac{d}{dt}f(g(t)) = f'(g(t)) g'(t).$$

If we write $y = f(x)$ and $x = g(t)$, then we have $y = f(x) = f(g(t))$, and so we can differentiate y with respect to t . The chain rule can then be written as

$$\frac{dy}{dt} = \frac{df}{dx} \frac{dx}{dt}.$$

If this looks a little bit strange, just realize that $\frac{df}{dx} = f'(x)$ and since $x = g(t)$, we have $\frac{df}{dx} = f'(x) = f'(g(t))$, and similarly $\frac{dx}{dt} = g'(t)$ since $x = g(t)$. So this really is the "usual" chain rule, just written in a way that might look a little bit foreign.

Now let's change things a little bit, let's suppose that f is a function of two variables, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. But let's also suppose that instead of plugging in (x, y) values into f , we plug in two different functions. Let's say that we take $x = g(t)$ and $y = h(t)$ and plug that into our function. This gives us a function of a single variable,

$$z(t) = f(x(t), y(t)).$$

Since this is a function of a single variable, we ought to be able to calculate its derivative, $\frac{dz}{dt}$. However, we have this multivariable function

$f(x, y)$ involved. How are the partial derivatives, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, related to the derivative $\frac{dz}{dt}$? The answer to this question is provided by the multivariable version of the chain rule.

Theorem 1.1 (Multivariable chain rule, version 1). *Suppose that $f(x, y)$ is a multivariable function whose partial derivatives exist and are continuous. Suppose also that $x = g(t), y = h(t)$ are two single variable functions whose derivatives exist and are continuous as well. The single variable function*

$$z(t) = f(g(t), y(t))$$

has the following derivative:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Proof. To show this, let's recall that we can write the derivative $\frac{dz}{dt}$ as the limit $\lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t}$. Now we note that the change in z , Δz , depends on the change in both x and y . That is, if we change our input x just a little bit, z must change as well; and likewise for y . We can thus write

$$\Delta z = \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y + \epsilon_1(\Delta x) + \epsilon_2(\Delta y).$$

The idea here is that the change Δz is approximately $\frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$. It may not be exact, though, so we add on some values $\epsilon_1(\Delta x)$ and $\epsilon_2(\Delta y)$ which “take up the slack” to make the equation exact. As Δx and Δy get small, these values $\epsilon_1(\Delta x)$ and $\epsilon_2(\Delta y)$ go to zero.

Now if we divide both sides of the above equation by Δt , we have

$$\frac{\Delta z}{\Delta t} = \frac{\partial f}{\partial x} \frac{\Delta x}{\Delta t} + \frac{\partial f}{\partial y} \frac{\Delta y}{\Delta t} + \frac{\epsilon_1(\Delta x)}{\Delta t} + \frac{\epsilon_2(\Delta y)}{\Delta t}.$$

Taking the limit as $\Delta t \rightarrow 0$ this becomes

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

(As $\Delta t \rightarrow 0$, we must have that $\epsilon_1(\Delta x) \rightarrow 0$ as well since $x = g(t)$. Similarly for y and $\epsilon_2(\Delta y)$.) $\square$

Example 1.1. Suppose that $z = f(x, y) = x^2 + y^2 + xy$ where $x = \sin(t)$ and $y = e^t$. Calculate $\frac{dz}{dt}$.

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \\ &= (2x + y) \cos(t) + (2y + x)e^t \\ &= (2\sin(t) + e^t) \cos(t) + (2e^t + \sin(t))e^t. \end{aligned}$$

Note that we could have first also replaced x and y with $\sin(t)$ and e^t , respectively, and then calculated the derivative:

$$\frac{dz}{dt} = \frac{d}{dt} (\sin^2(t) + e^{2t} + \sin(t)e^t).$$

This would give us the same derivative, but the advantage of the chain rule is that it lets us easily calculate the derivative, possibly shortcutting any necessary algebraic simplifications. In particular, if we repeated the above example keeping $f(x, y) = x^2 + y^2 + xy$, but changing x and y to some other functions, most of the work is already done. All we'd have to do is change our final answer.

Example 1.2. Suppose that $z = f(x, y) = x^2 + y^2 + xy$ where $x = \sqrt{t}$ and $y = \tan(t)$. Calculate $\frac{dz}{dt}$.

$$\begin{aligned} \frac{dz}{dt} &= (2x + y) \frac{1}{2\sqrt{t}} + (2y + x) \sec^2(t) \\ &= (2\sqrt{t} + \tan(t)) \frac{1}{2\sqrt{t}} + (2\tan(t) + \sqrt{t}) \sec^2(t). \end{aligned}$$

2. CHAIN RULE FOR COMPOSITION WITH A MULTIVARIABLE FUNCTION

Now suppose that we have $z = f(x, y)$, but instead of replacing x and y with functions of a single variable, we replace them with functions of two variables. Say $x = g(s, t)$ and $y = h(s, t)$. How z is a function of two variables,

$$z(s, t) = f(g(s, t), h(s, t)).$$

Since z is a function of two variables, we can discuss the partial derivatives of z with respect to either s or t . What does the chain rule look like in this situation?

If we wanted to calculate $\frac{\partial z}{\partial t}$, we'd differentiate like normal, treating s as a constant. This means that our chain rule above still applies, but we'd replace $\frac{dx}{dt}$ and $\frac{dy}{dt}$ with $\frac{\partial x}{\partial t}$ and $\frac{\partial y}{\partial t}$, respectively. We'd perform similar replacements if we wanted to calculate $\frac{\partial z}{\partial s}$, giving us the following version of the chain rule.

Theorem 2.1 (Multivariable chain rule, version 2). *Suppose that $f(x, y)$ is a multivariable function whose partial derivatives exist and are continuous. Suppose also that $x = g(s, t), y = h(s, t)$ are two multivariable functions whose partial derivatives exist and are continuous as well. The multivariable function*

$$z(s, t) = f(g(s, t), h(s, t))$$

has the following partial derivatives:

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}$$

Example 2.1. Suppose $z = f(\theta, \varphi) = \sin(\theta) \cos(\varphi)$ where we replace $\theta = st^2$ and $\varphi = t$. Calculate both of the first-order partial derivatives of z with respect to s and t .

$$\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial s} + \frac{\partial f}{\partial \varphi} \frac{\partial \varphi}{\partial s} \\ &= (\cos(\theta) \cos(\varphi)) \cdot t^2 - (\sin(\theta) \sin(\varphi)) \cdot 0 \\ &= \cos(st^2) \cos(t) t^2 \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial t} &= \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial t} + \frac{\partial f}{\partial \varphi} \frac{\partial \varphi}{\partial t} \\ &= (\cos(\theta) \cos(\varphi)) \cdot 2st - (\sin(\theta) \sin(\varphi)) \cdot 1 \\ &= 2st \cos(st^2) \cos(t) - \sin(st^2) \sin(t) \end{aligned}$$

3. IMPLICIT DIFFERENTIATION

We can use these ideas to reinterpret implicit differentiation for functions of a single variable.

Suppose that we have a curve in $\mathbb{R}^2$ given by the equation $y^2 = x^3 - x + 1$. See Figure 1.

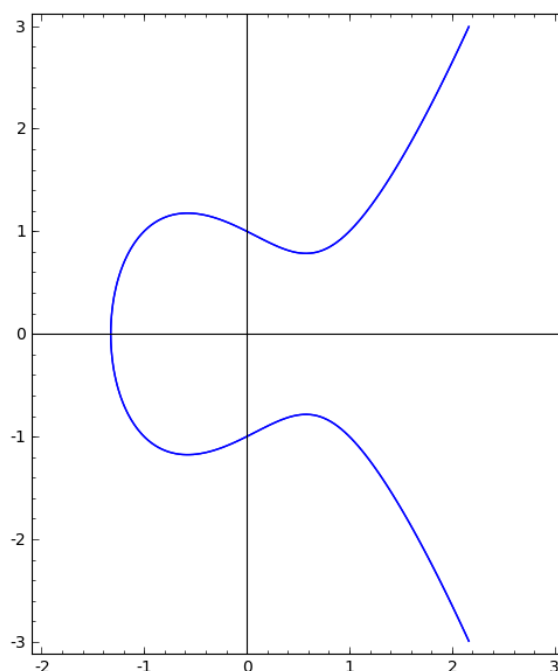
Notice that this curve isn't the graph of a function of x ; you can't solve the above equation for y (you'd get two possibilities for each x , a positive and a negative). However, it still makes sense to talk about slopes of lines tangent to this curve: we can still make sense of $\frac{dy}{dx}$, even though y isn't a function of x !

You of course already know how to do this with implicit differentiation from your first-semester calculus courses, but we can also view this as a multivariable problem. To do this, let's first rewrite our equation

$$y^2 = x^3 - x + 1$$

by putting everything on one side of the equation:

$$y^2 - x^3 + x - 1 = 0.$$

FIGURE 1. The curve determined by $y^2 = x^3 - x + 1$

Now consider the function $F(x, y) = y^2 - x^3 + x - 1$. Notice that our curve is the set of (x, y) points satisfying $F(x, y) = 0$.

Now, since we want to calculate $\frac{dy}{dx}$, let's suppose that y is a function of x – just call this $y(x)$. Then

$$\begin{aligned}
 & y^2 - x^3 + x - 1 \\
 \implies & F(x, y) = 0 \\
 \implies & \frac{d}{dx}F(x, y) = \frac{d}{dx}0 = 0 \\
 \implies & \frac{\partial F}{\partial x} \frac{dx}{dx} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0 \\
 \implies & F_x + F_y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{F_x}{F_y}.
 \end{aligned}$$

In our particular example, $F_x = -3x^2 + 1$ and $F_y = 2y$, hence

$$\frac{dy}{dx} = \frac{3x^2 - 1}{2y}.$$

This means the slope of the tangent line to $y^2 = x^3 - x + 1$ at the point $(2, \sqrt{7})$ on the curve is $\frac{5}{2\sqrt{7}}$.